# FINDING

# Agent stall root cause — my GOMAXPROCS attribution was WRONG

Captured 2026-08-07T10:10:12Z

## What I claimed

In commit 40c8d5ce and in ~8 agent briefs I stated the cgroup/GOMAXPROCS oversubscription was why agents overran the 600s stream watchdog, and told agents ‘this should materially reduce your chance of being killed’.

## Evidence 1 — transcript size INVERSELY predicts death

STALLED : 270, 470, 502, 552 KB mean ~449 COMPLETED: 499, 583, 685, 707, 806 KB mean ~656 Starvation predicts heavier work dies MORE. The opposite holds: the agents that did the most work completed. Size does not predict death.

## Evidence 2 — deaths do not cluster in time

stalls 2026-08-05 16:18, 17:17, 19:43, 08-06 16:58, 08-07 02:59 completes 2026-08-06 18:50, 21:03, 23:43, 08-07 05:57, 11:06 Interleaved, no clean separation, no common host condition at those moments.

## Evidence 3 — the decisive one: WHERE they died

All four stalled agents’ final text is an INCOMPLETE SENTENCE, composed just before issuing a tool call: HXC-229 ‘I’ll start by investigating the current state of the systemd unit’ HXC-218 ‘url.Parse is lenient here … let me check that recorded output’ HXC-217 ‘Now I have the full picture. Let me parallelize: dispatch’ HXC-221 ‘Fixing the fixture to be genuinely stateful:’

The watchdog measures STREAM output. A CPU-starved agent still emits tokens, only slower, and would die DURING a long command with that command as its last action. These died while COMPOSING TEXT — the token stream stopped midgeneration and never resumed. That is an API/stream-layer condition, not a host-resource one.

## Corrected conclusion

The cgroup finding is REAL and stands on its own: cpu.max 860000/100000 = 8.6 CPUs vs nproc 64, 50.3% of periods throttled, 75.5h frozen, GOMAXPROCS=8 beat 64 on a trivial build. The Makefile fix is correct and worth keeping.

But it does NOT explain the stalls, and I asserted that it did without testing it. Two separate facts got welded into one causal story because they appeared in the same investigation.

## What actually mitigates stalls

Respawn-on-stall with preserved partial work — which is what SS11.4.147 already prescribes and what has in fact recovered every stalled agent today. Not a tuning knob. The mitigation was already correct; my explanation was not.

## Honest boundary

I cannot fix an upstream token-stream interruption from here. What I can do, and have done, is stop telling agents a CPU setting will protect them.